32. A probability distribution of a random variable X is given by

X	–2	3	1
P (x)	$\frac{1}{3}$	$\frac{1}{2}$	$\frac{1}{6}$

Find $E(2X+5)$

33. If three coins are tossed find the mean and variance of the number of heads.

34. A discrete random variable has the following distribution function

X	0	1	2	3	4	5	6	7	8
P(x)	a	$3a$	$5a$	$7a$	$9a$	$11a$	$13a$	$15a$	$17a$

Find (i) a (ii) $P(X<3)$ (iii) $P(X\geq 5)$ (iv) $P(3<X<7)$

35. Two cards are drawn with replacement from a well shuffled pack of 52 cards. Find the mean and variance of the number of Aces.

36. In an entrance examination student has answered all the 120 questions. Each question has 4 options and only one option is correct. A student gets one mark for correct answer and looses $\frac{1}{2}$ marks for wrong answer. What is the expectation of a mark scored by a student if he chooses the answer to each question at random?

37. A box contains 8 items of which 2 are defective. A man selects 3 items at random. Find the expected number of defective items he has drawn.

38. A box contains 4 white and 3 red balls. Find the probability distribution of number of red balls in 3 draws one by one from the box.

(i) With replacement (ii) Without replacement

39. The joint probability function of X and Y is given below:

X \ Y	1	3	9
2	0.1	0.1	0.05
4	0.2	0	0.1
6	0.1	0.15	K

Find (i) K (ii) $E(X+Y)$